



UNSW
SYDNEY

MATH1231 Mathematics 1B – Algebra

Section 08 - Linear Transformations as Matrices & Geometric Examples.

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Slides: David Angell, Sean Gardiner, Laure Helme-Guizon and others

Maps and matrices

Matrices and linear transformations. There is a very close relationship between linear transformations and matrix multiplication.

- We have already seen that if A is an $m \times n$ matrix and $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ is defined by $T(\vec{x}) = A\vec{x}$, then T is linear.

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and so on: the k th column of an $m \times n$ matrix A is $A\mathbf{e}_k$, where $\mathbf{e}_k$ is the k th standard basis vector in $\mathbb{R}^n$. So if T is known and we want to find the matrix A with $T(\vec{x}) = A\vec{x}$, we have

$$\{\text{\textit{k}th column of } A\} = A\mathbf{e}_k = T(\mathbf{e}_k).$$

Linear map are matrix multiplication “in disguise”

Theorem (**Matrix representation of linear maps**)

Let $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ be a linear transformation; for $k = 1, 2, \dots, n$, let $\mathbf{e}_k$ be the k th standard basis vector in $\mathbb{R}^n$. If A is the matrix whose k th column is $T(\mathbf{e}_k)$, then

$$T(\vec{\mathbf{x}}) = A\vec{\mathbf{x}} \quad \text{for every } \vec{\mathbf{x}} \text{ in } \mathbb{R}^n,$$

where both $\vec{\mathbf{x}}$ and $T(\vec{\mathbf{x}})$ are written as column vectors.

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Theorem (Matrix representation of linear maps)

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where both $\vec{x}$ and $T(\vec{x})$ are written as column vectors.

Proof. Writing $\vec{x}$ as $(x_1, x_2, \dots, x_n)^T$ and using the definition of matrix multiplication,

$$A\vec{x} = x_1 \{\text{1st column of } A\} + x_2 \{\text{2nd column of } A\} + \dots + x_n \{\text{nth column of } A\}$$

Since T is linear, we have

$$T(\vec{x}) = T(x_1\mathbf{e}_1 + x_2\mathbf{e}_2 + \dots + x_n\mathbf{e}_n) = x_1 T(\mathbf{e}_1) + x_2 T(\mathbf{e}_2) + \dots + x_n T(\mathbf{e}_n) = A\vec{x}.$$

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TIP!

Note that the theorem not only tells you that it is possible to rewrite T as a multiplication by a matrix A , but also it tells you how to construct the matrix A :

$$A = (T(\mathbf{e}_1) \mid T(\mathbf{e}_2) \mid \dots \mid T(\mathbf{e}_n))$$

is the matrix whose columns are the images of the standard basis vectors under T .

Example 1. Consider the linear map $T : \mathbb{R}^3 \rightarrow \mathbb{R}^4$ defined by

$$T \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} x_1 + 2x_2 + x_3 \\ -2x_1 + x_2 \\ 3x_2 - x_3 \\ -x_1 + 4x_3 \end{pmatrix}. \text{ Find a matrix } A \text{ such that } T(\vec{x}) = A\vec{x} \text{ for all } \vec{x} \in \mathbb{R}^3.$$

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$$T(\mathbf{e}_1) = T \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ -2 \\ 0 \\ -1 \end{pmatrix}, \quad T(\mathbf{e}_2) = T \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 2 \\ 1 \\ 3 \\ 0 \end{pmatrix}, \quad T(\mathbf{e}_3) = T \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \\ -1 \\ 4 \end{pmatrix},$$

so the matrix

Example 7. Consider the linear map $T : \mathbb{R}^3 \rightarrow \mathbb{R}^4$ defined by

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so the matrix

$$A = \begin{pmatrix} 1 & 2 & 1 \\ -2 & 1 & 0 \\ 0 & 3 & -1 \\ -1 & 0 & 4 \end{pmatrix}$$

has the property that $T(\vec{x}) = A\vec{x}$ for all $\vec{x} \in \mathbb{R}^3$.

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Alternative method: just write out

$$\begin{pmatrix} x_1 + 2x_2 + x_3 \\ -2x_1 + x_2 \\ 3x_2 - x_3 \\ -x_1 + 4x_3 \end{pmatrix} = \begin{pmatrix} & & \\ & & \\ & & \\ & & \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}$$

and fill in the blanks in the matrix!

Differentiation of quadratics can be written as a matrix!

Comment. Although the above example only shows how to write a linear transformation T in terms of matrix multiplication if the domain and codomain of T are $\mathbb{R}^n$ and $\mathbb{R}^m$, something similar can be done in other vector spaces. For example, consider the differentiation map

$$T : \mathbb{P}_2 \rightarrow \mathbb{P}_1 \quad \text{where} \quad T(p) = p' = \frac{dp}{dx}.$$

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$$T(a + bx + cx^2) = b + 2cx ;$$

if we write the polynomials as column vectors, we have

$$T \begin{pmatrix} a \\ b \\ c \end{pmatrix} = \begin{pmatrix} b \\ 2c \end{pmatrix} = \begin{pmatrix} \\ \\ \end{pmatrix} \begin{pmatrix} a \\ b \\ c \end{pmatrix} = A \begin{pmatrix} a \\ b \\ c \end{pmatrix}, \quad (*)$$

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$$A = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 2 \end{pmatrix}.$$

Differentiation can be written as a matrix, continued.

For example,

$$\begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 2 \end{pmatrix} \begin{pmatrix} 3 \\ -4 \\ 5 \end{pmatrix} = \begin{pmatrix} -4 \\ 10 \end{pmatrix}$$

is just another way of saying

$$\frac{d}{dx}(3 - 4x + 5x^2) =$$

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To know more about this, read section 7.6 of the course notes (in the course pack on Moodle).

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Warning. Don't take equation (*) too literally! In order to give a simplified rough idea of the topic, we have skipped over some important details. In particular, make sure you are clear that the polynomial and vector

$$a + bx + cx^2 \quad \text{and} \quad \begin{pmatrix} a \\ b \\ c \end{pmatrix}$$

are **not** the same thing.

Geometric examples of linear transformations



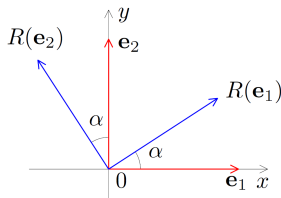
Geometric examples of linear transformations. Many important examples of linear mappings on $\mathbb{R}^2$ have simple geometric interpretations. These include stretching, compression, dot products and projections, which are explained in the notes. Here are some further examples. Our main aim will be to **find the matrix of each of these linear transformations.**

Rotations as linear maps represented by matrices

Rotations . Let α be a fixed angle, and for any vector $\vec{x}$ in $\mathbb{R}^2$, let $R(\vec{x})$ be the vector obtained by rotating $\vec{x}$ through an angle α about the origin. In example 2 we discussed the case when α is a right angle, and explained geometrically why R is linear; the proof for any α is essentially identical.

From the diagram we can see that

and so R is represented by the matrix



Matrix representation of linear maps. Let $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ be a linear transformation; for $k = 1, 2, \dots, n$, let $\mathbf{e}_k$ be the k th standard basis vector in $\mathbb{R}^n$. If A is the matrix whose k th column is $T(\mathbf{e}_k)$, i.e. $A = (T(\mathbf{e}_1) \mid T(\mathbf{e}_2) \mid \dots \mid T(\mathbf{e}_n))$ then

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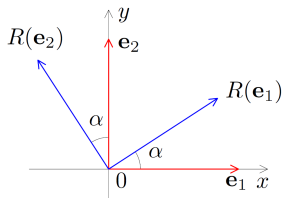
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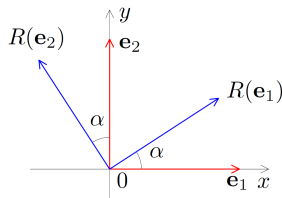
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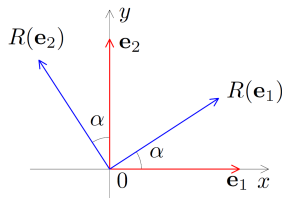
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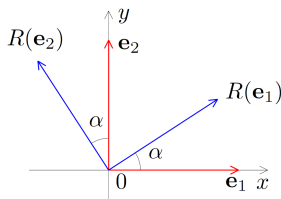
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Rotations as linear maps represented by matrices

Exercise 9. Find the vector $R(\vec{u})$ obtained by rotating $\vec{u} = \begin{pmatrix} 4 \\ -2 \end{pmatrix}$ through an angle of 30° .

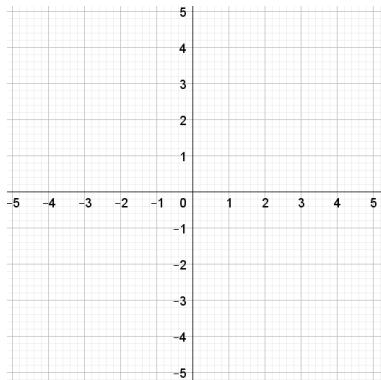
Solution. You could do this by drawing a careful diagram and using congruent triangles, but it would be messy. However, it is easy to get the answer by ...

Rotations as linear maps represented by matrices

Exercise 9. Find the vector $R(\vec{u})$ obtained by rotating $\vec{u} = \begin{pmatrix} 4 \\ -2 \end{pmatrix}$ through an angle of 30° .

Solution. You could do this by drawing a careful diagram and using congruent triangles, but it would be messy. However, it is easy to get the answer by ...using the above matrix

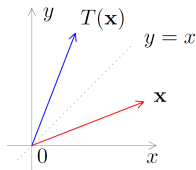
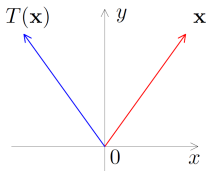
$$\begin{aligned} R(\vec{u}) &= \begin{pmatrix} \cos 30^\circ & -\sin 30^\circ \\ \sin 30^\circ & \cos 30^\circ \end{pmatrix} \begin{pmatrix} 4 \\ -2 \end{pmatrix} \\ &= \begin{pmatrix} \frac{\sqrt{3}}{2} & -\frac{1}{2} \\ \frac{1}{2} & \frac{\sqrt{3}}{2} \end{pmatrix} \begin{pmatrix} 4 \\ -2 \end{pmatrix} \\ &= \begin{pmatrix} 2\sqrt{3} + 1 \\ 2 - \sqrt{3} \end{pmatrix} \approx \begin{pmatrix} 4.5 \\ 0.3 \end{pmatrix}. \end{aligned}$$



Matrix representation of linear maps. Let $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ be a linear transformation; If $A = (T(\mathbf{e}_1) \mid T(\mathbf{e}_2) \mid \dots \mid T(\mathbf{e}_n))$ then $T(\vec{x}) = A\vec{x}$ for every $\vec{x}$ in $\mathbb{R}^n$.

Reflections, translations

Reflections. The following diagrams illustrate a reflection in the y -axis, and a reflection in the line $y = x$.

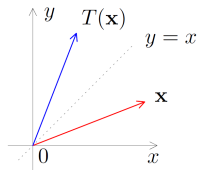
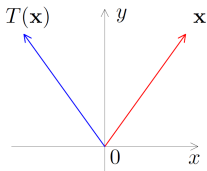


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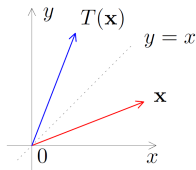
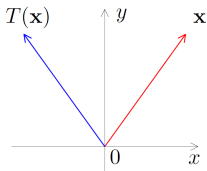


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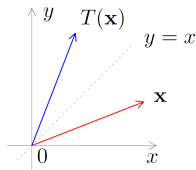
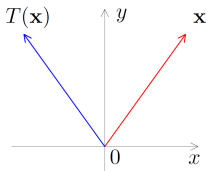
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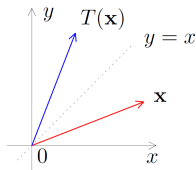
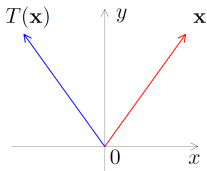
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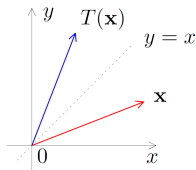
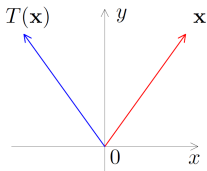
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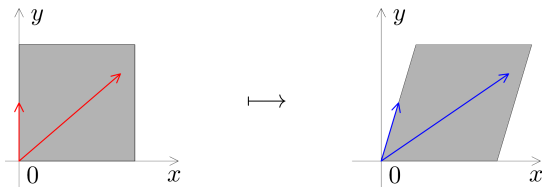
Translations.



A translation is **not** a linear transformation as it does not map the origin to the origin, that is, $T(\vec{\mathbf{0}}) \neq \vec{\mathbf{0}}$.

Shear

A **shear mapping parallel to the x -axis** can be visualised by imagining a stack of paper with sheets lying horizontally being pushed so that the vertical edge becomes oblique.



For any point (x, y) , this mapping leaves the y -coordinate unchanged, and increases the x -coordinate by an amount proportional to the y -coordinate. Therefore

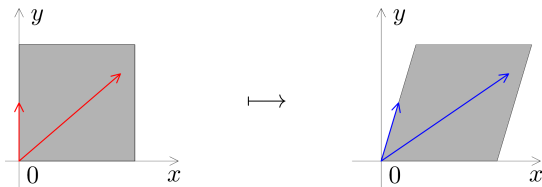
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$$A = \begin{pmatrix} 1 & k \\ 0 & 1 \end{pmatrix}.$$

Exercise 10. Find the matrix of a shear mapping parallel to the y -axis.

linear transformation map lines to lines (or points)

The idea: An important feature of all these mappings is that the image of a line under the transformation is also a line. This is always true for linear mappings, except for certain cases in which the direction of the line is “reduced to nothing” and so we obtain only a point.

Fact: A linear transformation maps a line to a line (or a point)

Let $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ be linear and let ℓ be a line in $\mathbb{R}^n$. Then the image of ℓ under T , that is,

$$\{ T(\vec{x}) \mid \vec{x} \text{ is on } \ell \} ,$$

is either a line or a point in $\mathbb{R}^m$.

Proof. We know from MATH1131 that we can write the line as

$$\ell = \{ \vec{a} + \lambda \vec{v} \mid \lambda \in \mathbb{R} \}$$

for certain fixed vectors $\vec{a}$ and $\vec{v}$. The image of this line is

$$\{ T(\vec{a} + \lambda \vec{v}) \mid \lambda \in \mathbb{R} \} = \{ T(\vec{a}) + \lambda T(\vec{v}) \mid \lambda \in \mathbb{R} \} .$$

Now $T(\vec{a})$ and $T(\vec{v})$ are fixed vectors; if $T(\vec{v}) = \vec{0}$ then the set we have just written down consists of the single vector $T(\vec{a})$, otherwise it is a line through $T(\vec{a})$ in the direction $T(\vec{v})$.